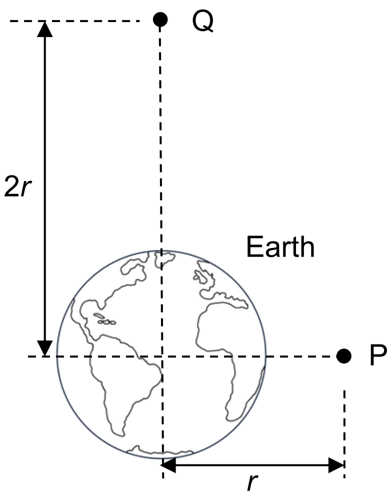
10 P and Q are two points above Earth's surface at distances r and $2r$ respectively from the centre of the Earth.



The gravitational potential at P is -800 kJ kg^{-1}. What is the gain in gravitational potential energy of a 2 kg mass when it is moved from P to Q?

A	- 400 kJ	**B**	- 200 kJ	**C**	400 kJ	**D**	800 kJ

Answer: D

Since the distance from earth changes from r to 2r, the gravitational potential at Q is -400 kJ kg^{-1}.

Gain in gravitational potential energy
= final GPE – initial GPE
= 2.00 (-400 – (-800))
= 800 kJ

11 Planet X has a density of ρ and radius of R. The gravitational acceleration at its surface is g